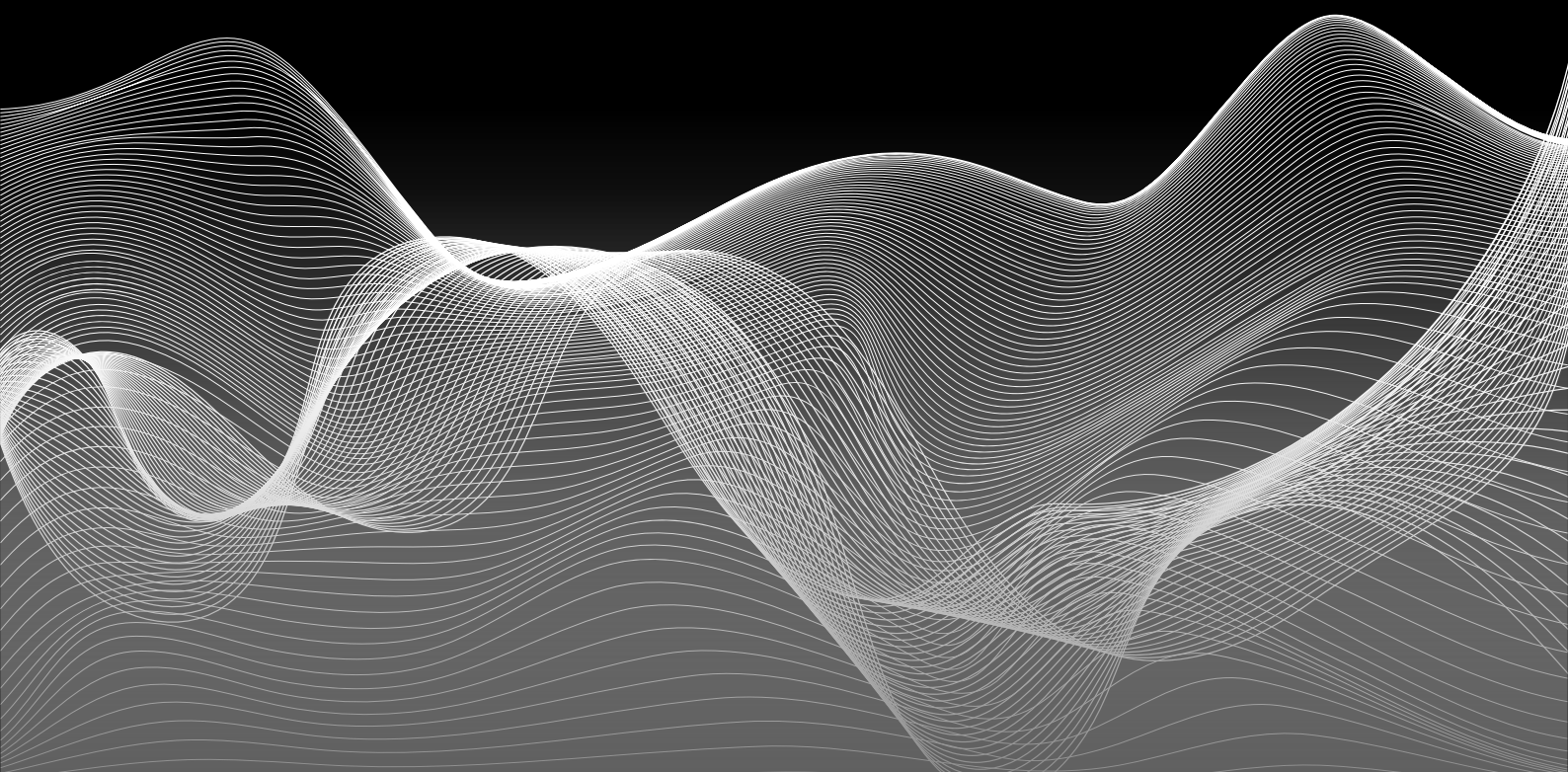
BIOMEDICAL ENGINEER

AMNA KHAN
PORTFOLIO





AMNA KHAN

Biomedical Engineer

EMPOWERING HEALTHCARE THROUGH ENGINEERING

Hello! Sending you greetings and positive vibes from Turkey, where this Pakistani girl from Rawalpindi landed with a Turkiye Burslari Scholarship and eyes full of dreams to create something extraordinary. My journey has taken me from mastering the Turkish language to building friendships across continents and yes, discovering that I'm absolutely obsessed with exploring every cuisine imaginable. From Turkish delights to Pakistani karahi, good food has become my universal language for connecting with people.

I'm a biomedical engineer who believes in the magic of turning pain into possibility. My mission is to lead innovativesolutions that make people's lives better, one breakthrough at a time. I specialize in rallying diverse teams around shared visions, whether I'm pioneering new approaches to healthcare challenges or mentoring others to push boundaries alongside me. This cross-cultural journey has taught me that the most powerful innovations happen when different perspectives unite around a common purpose. I bring that same collaborative leadership to every project building bridges between disciplines, cultures, and complex problems to create meaningful impact. This portfolio captures the milestones, breakthroughs, and connections that define both who I am and the future I'm actively building.

Adana, Türkiye | LinkedIn | (+90)5056687366 | amna.khan92713@gmail.com

MY RESEARCH JOURNEY

FROM BLUEPRINTS TO BREAKTHROUGHS

PUBLICATIONS & RESEARCH

- Paper 1: AI-Based System for Minimizing Radiation Risk and Early Lung Cancer Detection
- Paper 2: Multimodal Bio signal Fusion for Real-Time Stress Detection: A Comparative Study using Machine Learning Algorithms for Enhanced Predictive Accuracy
- Paper 3: Designing of an automated ablution device through implementation of Universal Design Principles

MEDICAL IMAGING & BIOSIGNAL PROCESSING

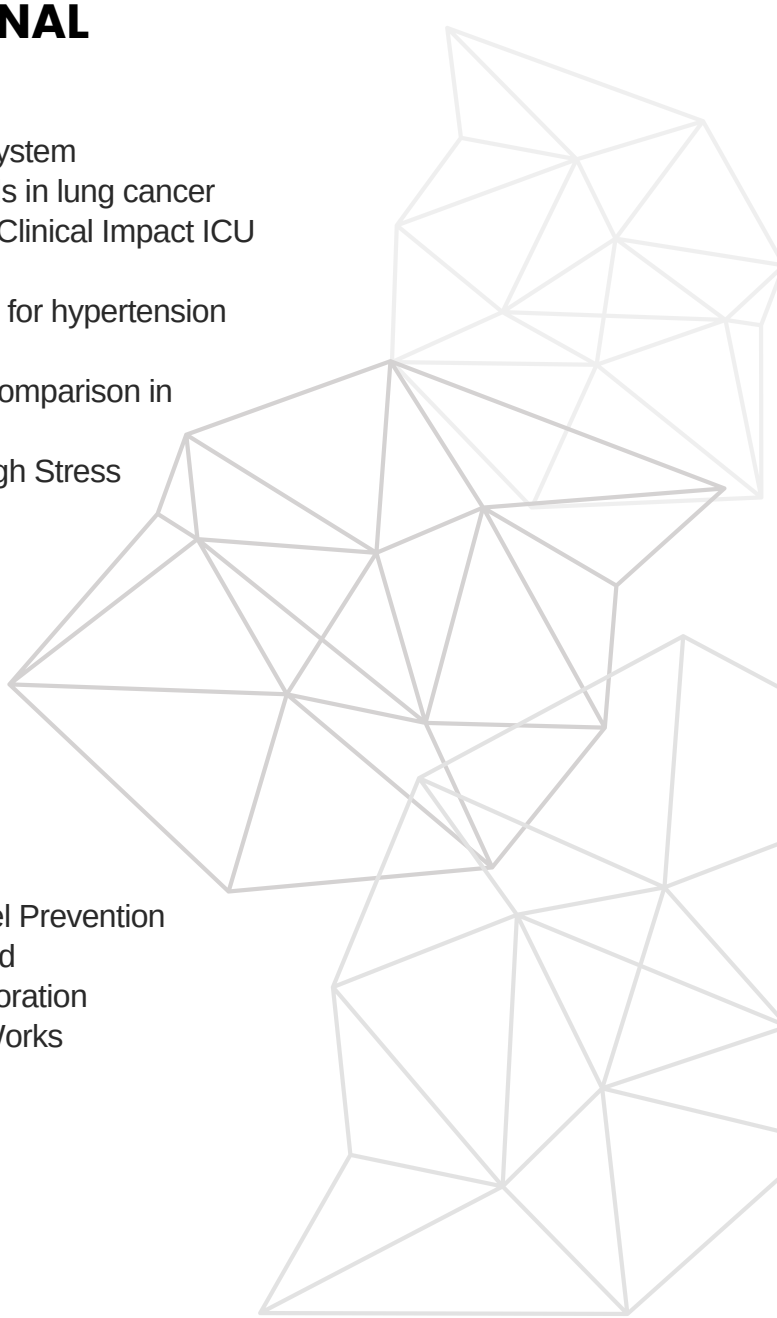
- Project 1: 3D CT Early-stage Cancer Detection System
- Project 2: Bibliometric analysis for research trends in lung cancer
- Project 4: Translating Reinforcement learning for Clinical Impact ICU patients hypotension dose management
- Project 5: Device Synchronization, data collection for hypertension detection and Mentorship
- Project 6: Multi institutional stress detection and comparison in school children
- Project 7: Measurement of stress reduction through Stress Management Games

ELECTRONICS

- Project 3: From Campus to Global Summit: The SPIROCARE+ Journey
- Project 8: Cost-Effective Spirometer Prototype

MECHANICS & DESIGN

- Project 9: Ergonomic Monitoring for Carpal Tunnel Prevention
- Project 10: Automated Ablution device for disabled Mechanical Design and Cross-disciplinary Collaboration
- Project 11: 3D Prosthetic Finger Design in SolidWorks



EXPLORING. LEADING. CREATING



PUBLICATION 1

AI-BASED SYSTEM FOR MINIMIZING RADIATION RISK AND EARLY LUNG CANCER DETECTION

- **Details:** 9. Uluslararası Öğrenci Bilimler Kongresi, 2025 Bursa Turkey (Accepted paper)
- **Related Projects:** See Project 1 & 2 for details
- **Abstract:** Lung diseases remain a pressing global health challenge, requiring both scientific insight into research evolution and practical advances in diagnostic solutions. To address this, the present study integrates bibliometric analysis with experimental validation of computational models, offering a combined medical and technical perspective. Research articles were retrieved from Web of Science Core Collection, Scopus, PubMed, Dimensions.ai, IEEE Xplore, and Google Scholar, and systematically analyzed using VOSviewer and CiteSpace to identify publication trajectories, research hotspots, and institutional collaborations. Drawing upon these bibliometric insights, deep learning models for three-dimensional lung nodule detection were developed and applied to low-dose CT scans obtained from 605 cancer patients. Model performance was assessed using sensitivity, Dice coefficient, and false positive rates. The bibliometric results indicated sustained growth in lung disease research, highlighting radiomics and AI-driven methodologies as emerging focal points, alongside significant international cooperation. On the experimental side, the models achieved a peak Dice score of 0.9220, demonstrating strong sensitivity for small nodules and reduced false detections when compared with conventional baselines. By bridging large-scale research mapping with empirical validation on real-world patient data, this study underscores both the scientific directions shaping lung disease research and the clinical utility of advanced computational approaches, thereby providing valuable guidance to researchers, technologists, and healthcare practitioners alike.
- **Keywords:** Bibliometric analysis, low-dose CT scan, Lung Cancer, Deep Learning Models, Artificial İntelligence, Artificial Intelligence İn Healthcare, Cancer and Treatments, Interventional Radiology, Diagnostic Medicine

PUBLICATION 2

MULTIMODAL BIO SIGNAL FUSION FOR REAL-TIME STRESS DETECTION: A COMPARATIVE STUDY USING MACHINE LEARNING ALGORITHMS FOR ENHANCED PREDICTIVE ACCURACY

- **Details:** 8. Uluslararası Öğrenci Bilimler Kongresi, 2024 Gaziantep, Turkey (Book Chapter)
- **Related Projects:** See Project 6 for details
- **Abstract:** In the past ten years, “physiological stress detection” has become an increasingly important topic. In many cases stress in chronic form results in health problems including cancer, cardiovascular disease, depression, and diabetes. These physiological diseases along with several psychological problems can be easily prevented through the detection of stress at early stages. Previously numerous researchers have employed bio signals for stress level estimation. Most of these works utilize one, two or a maximum of three bio signals simultaneously but this study includes multi bio signals, where acquisition of six different bio signals including EEG, EMG, EDA, PPG, respiration and temperature signals took place. This helped in providing insights into central stress responses through EEG and EMG, along with peripheral stress responses through PPG, EDA, temperature, contributing to a more comprehensive stress analysis. This resulted in enhancing the accuracy, reliability, and depth of understanding of stress-related responses. The protocol was designed in collaboration with a behavioral medicine specialist, ensuring activity of regions that are responsible for cognitive control and executive functioning is well monitored employing the Stroop Test, while the exercise sessions tested the relation of exercise to motor control regions and areas linked to coordination and movement. Real time stress level monitoring was done for students in a classroom environment, at numerous elementary Schools. The acquired bio signals were processed to extract useful information using MATLAB, necessary processing was done, and machine learning models were implemented to detect the level of stress experienced by individuals. The results with 90% accuracy, were able to differentiate stress and non-stress responses. Furthermore, classification provided statistically significant results in estimation of stress level. Hence, it can be indicated that this multimodal bio signal fusion to be an accurate source of stress detection in future. This study will act as a stepping stone for the researchers and engineers interested in the development of multi mode bio signal fusion stress detection systems as well as the signal processing engineers to improve the algorithms for stress detection models enhancement.
- **Keywords:** Stress Detection, Medical Signal Processing, Machine Learning Algorithms, Real-Time Monitoring Systems, Cross-Signal Analysis

PUBLICATION 3

DESIGNING OF AN AUTOMATED ABLUTION DEVICE THROUGH THE IMPLEMENTATION OF UNIVERSAL DESIGN PRINCIPLES

- **Details:** 7. Uluslararası Öğrenci Bilimler Kongresi, 2023 Istanbul Turkey (Book Chapter)
- **Related Projects:** See Project 10 for details
- **Abstract:** Ablution, the act of cleaning oneself before performing prayers. However, this process requires repetitive motion of various parts of body and the disabled people, especially the elderly, face challenges while performing this ritual by themselves. This paper introduces an ablution device, a machine that combines practicality, accessibility, and water-saving characteristics. The tool accommodates people with various body shapes, heights, and mobility disabilities by applying universal design principles. The inclusiveness is ensured by the ergonomic design, which tackles upper body constraints. Portability and sensory accessibility were also taken into consideration, making it usable for individuals having sensory limitations. Water-saving techniques are used to encourage conservation. With the help of Auto-CAD a three-dimensional model of the machine was also produced. The design effectively prompts and offers feedback to help with task completion. The implementation of this device on a bigger scale could enable people with disabilities to perform ablution with ease and save water.
- **Keywords:** ablution, portable system, ergonomics ablution, water conservation, universal design

LINK FOR PUBLICATIONS:

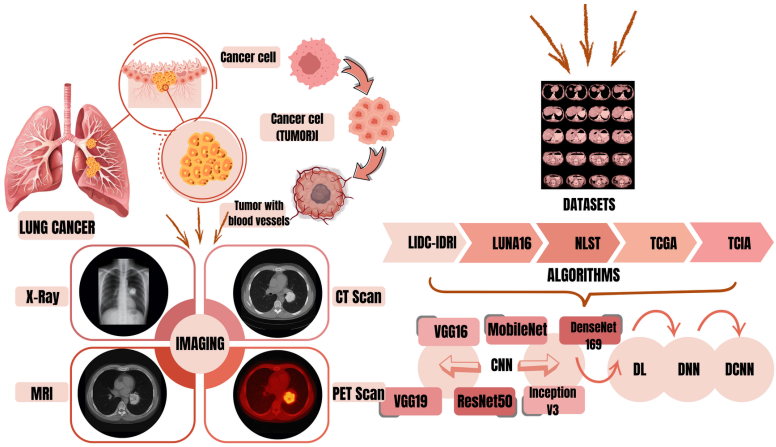
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PROJECT 1

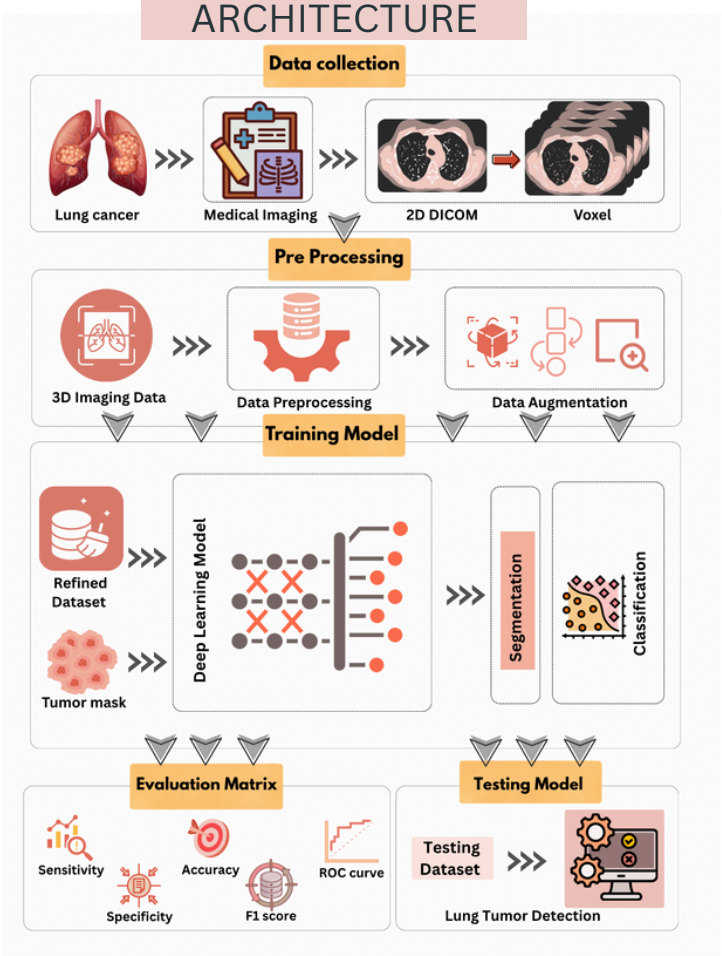
3D CT EARLY STAGE LUNG CANCER DETECTION SYSTEM

- **Challenge:** Radiologists needed a reliable system for early lung cancer detection, but existing tools are not designed for low dose CT and clinical integration.
- **Action:** Collaborated with radiologists and engineers, translated diagnostic workflows into technical requirements, designed system architecture using 3D volume of low dose CT and performed segmentation of cancer.
- **Result:** Delivered a clinically aligned system that achieved dice score of 0.798.

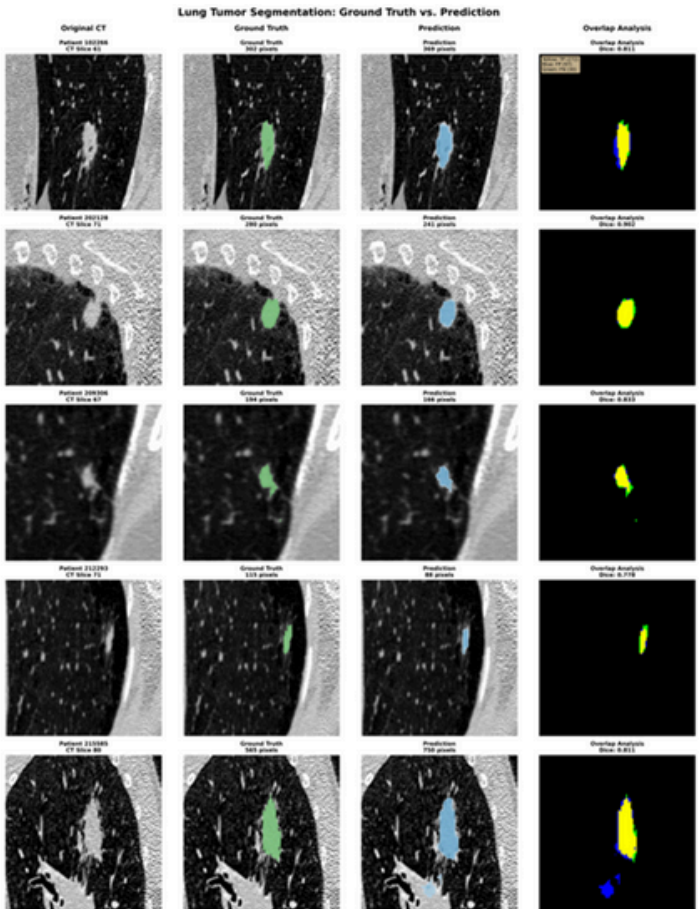
THEORETICAL FRAMEWORK



RESEARCH ARCHITECTURE

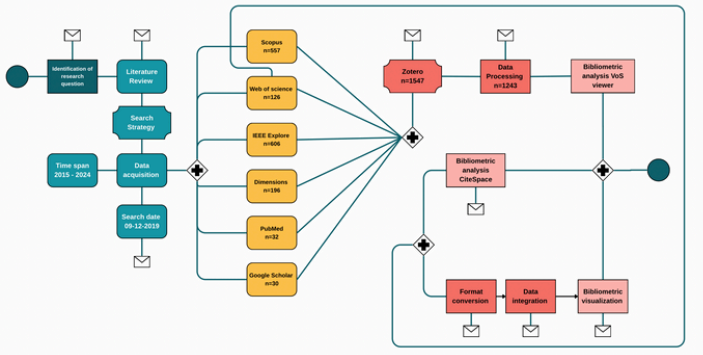


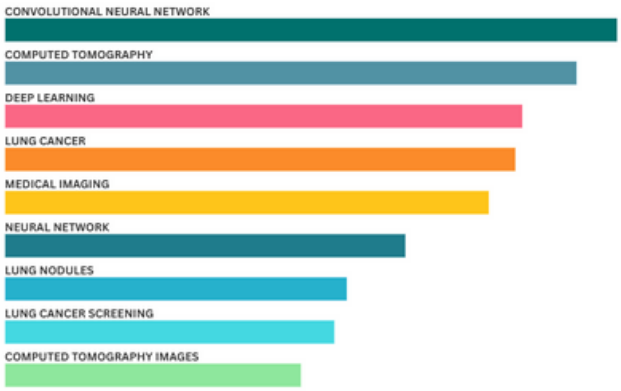
LUNG CANCER SEGMENTATION



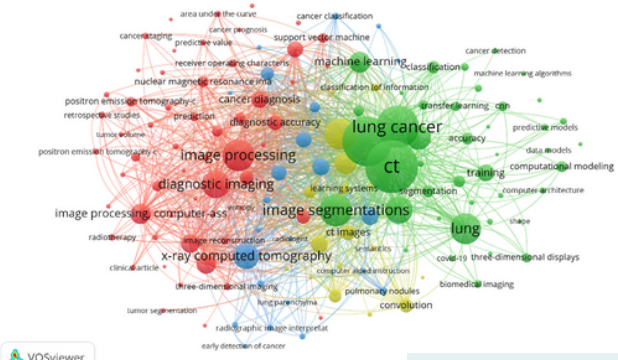
PROJECT 2

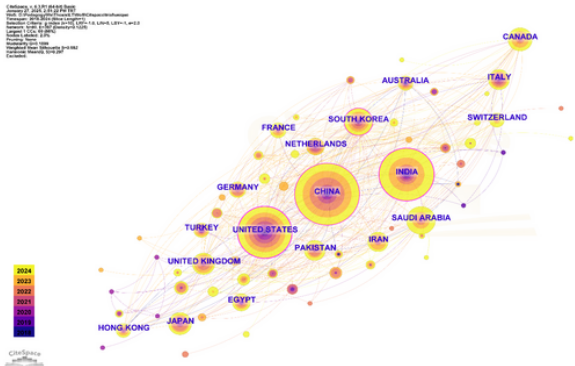
BIBLIOMETRIC ANALYSIS FOR RESEARCH TRENDS IN LUNG CANCER





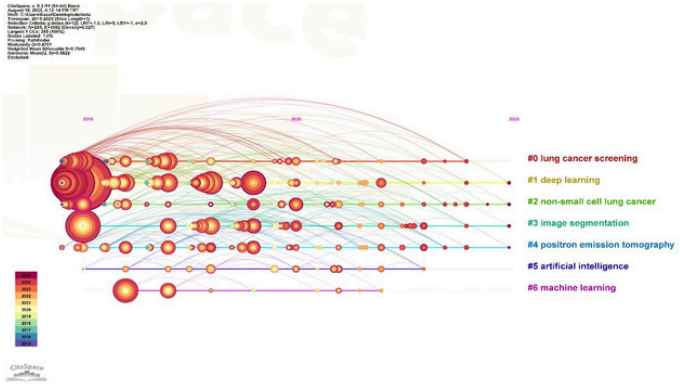
- **Challenge**: The rapid growth of research in lung cancer medical imaging and AI makes it difficult to track emerging trends, key contributors, and collaborative networks.
- **Action:** I conducted a bibliometric analysis using six major databases, cleaning and merging the data, then analyzing publication trends, keywords, co-author networks, institutions, and countries with VOSviewer and CiteSpace to identify hotspots in AI and imaging research.
- **Result:** The analysis revealed increasing publications in AI-driven lung cancer imaging, with key focus areas and research gaps were identified. Influential authors, institutions, and journals were identified, highlighting global collaborations and guiding future research directions. This bibliometric analysis laid foundation for the early stage lung cancer detection project.





CO-OCCURRENCE ANALYSIS



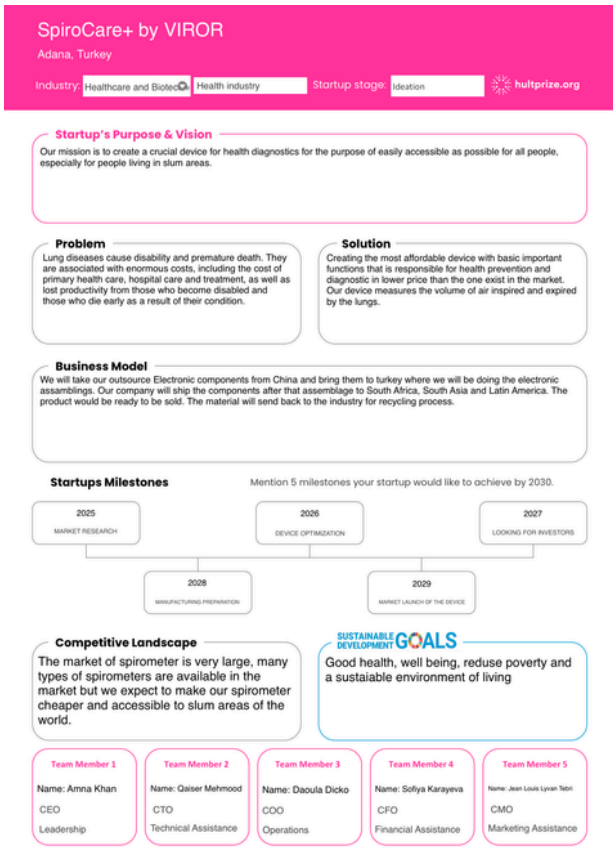


CO-AUTHORSHIP ANALYSIS

PROJECT 3

FROM CAMPUS TO GLOBAL SUMMIT: THE SPIROCARE+ JOURNEY

- **Challenge**: Address the critical gap in respiratory healthcare diagnostics in underserved communities where traditional spirometers are too expensive, preventing early detection and management of respiratory diseases.
- **Action:** Led a multicultural team from Pakistan, Kazakhstan, Ivory Coast, and Mali as CEO and Team Lead, bringing together diverse perspectives to develop an affordable spirometer solution. Managed product development strategy and coordinated cross-cultural collaboration to create a unified vision. Secured funding from sponsors to enable the team's participation at the Global Summit. Presented the business case and medical technology solution to judges and potential investors through compelling pitch presentations.
- **Result:** Won the university-level Hult Prize competition and secured funding to advance to the international stage. Selected to present at the Hult Prize Global Summit, demonstrating effective cross-cultural leadership and social innovation in healthcare.



STARTUP PITCH LINK



ÇUKUROVA ÜNİVERSİTESİ
ON- CAMPUS HULT PRIZE WINNER

viror
LIFE IS BREATHING
+
YURTDIŞI TÜRKLER VE AKRABA TOPLULUKLAR BAŞKANLIĞI
ÇUDER
ÇUKUROVA ULUSLARARASI ÖĞRENCİ DERNEĞİ
ÇUKUROVA ÜNİVERSİTESİ
TÖMER

SUMMIT FUNDING & SPONSORSHIP





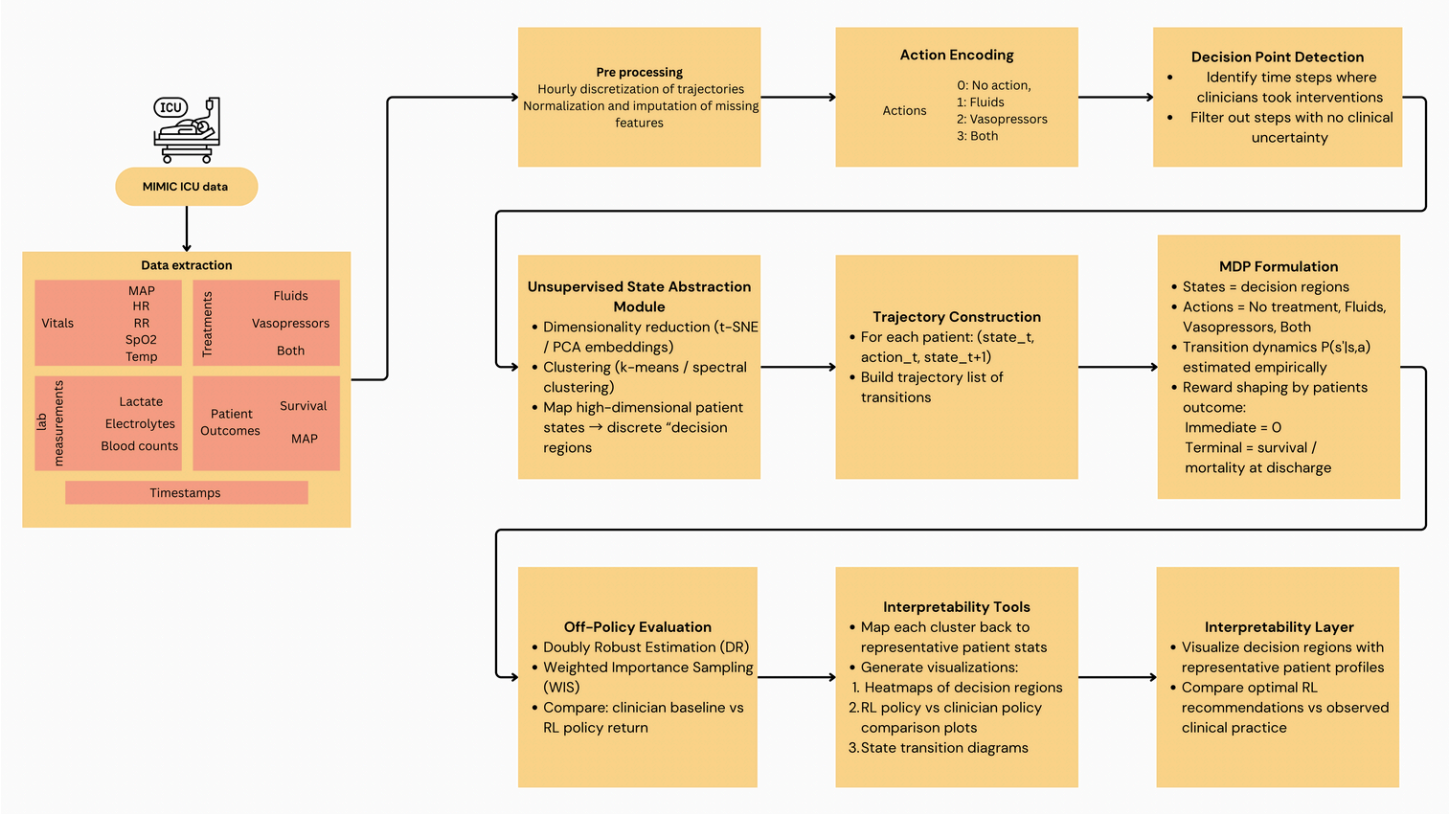
DUBAI GLOBAL SUMMIT HULT PRIZE

PROJECT 4

CASE STUDY: TRANSLATING REINFORCMENT LEARNING FOR CLINICAL IMPACT HYPOTENTION MEDICINE IN ICU

- **Challenge:** Reinforcement learning offers breakthrough potential for ICU care, but technical complexity creates barriers between AI engineers and clinical teams. A sophisticated RL framework for hypotension treatment (Pan et al., 2022, npj Digital Medicine) exemplified this gap –advanced code that could guide critical decisions, yet difficult to interpret for hospital implementation.
- **Action:**
  1. Reverse-engineered the complete GitHub repository, tracing each component from data preprocessing through value iteration algorithms
  2. Decoded the Markov Decision Process to understand how technical operations mapped to clinical reasoning
  3. Created workflow diagrams and annotated code explanations, translating RL concepts into plain language
  4. Built visual frameworks connecting computational logic to bedside decision-making
- **Result:** Demonstrated the rare ability to independently master complex AI implementations and translate them for clinical application–the essential bridge hospitals need between data science teams and medical practice. This work showcases capability to decode engineers' technical solutions, evaluate their clinical feasibility, and lead interdisciplinary research where communication typically breaks down.

Research Architecture



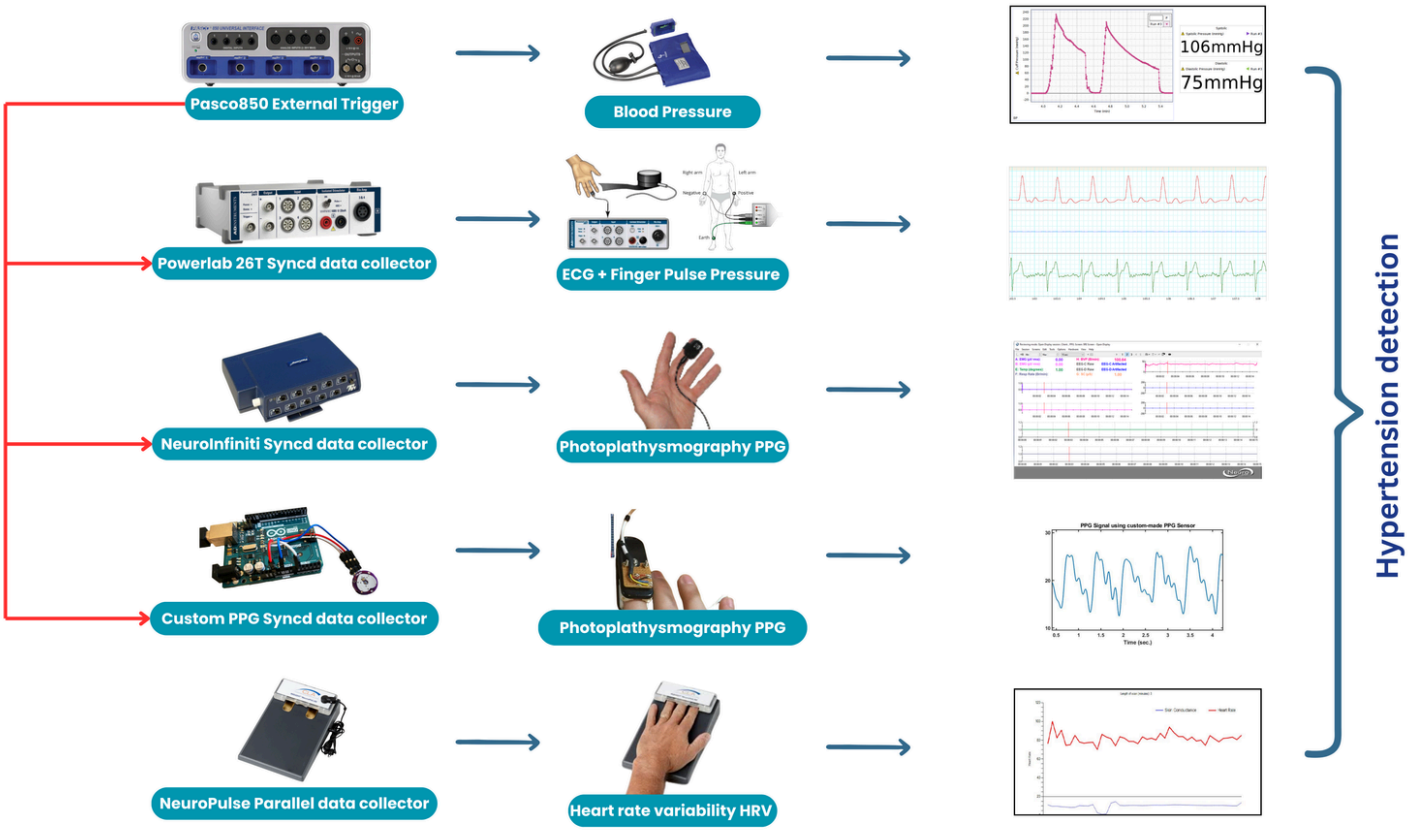
PROJECT 5

DEVICE SYNCHRONIZATION, DATA COLLECTION FOR HYPERTENSION DETECTION AND MENTORSHIP



- **Challenge:** Research lab at Riphah International University faced inefficiencies due to unsynchronized diagnostic devices, causing workflow delays and inconsistent data.
- **Action:** Designed research protocols, synchronized bio signal acquisition systems including PASCO, PowerLab, NuroInfiniti, Custom PPG and NeuroPulse for simultaneous data acquisition, trained engineers, and mentored a team of five while leading study protocol design.
- **Result:** Achieved consistent data collection, reduced workflow delays, and built long-term research capacity through mentorship.

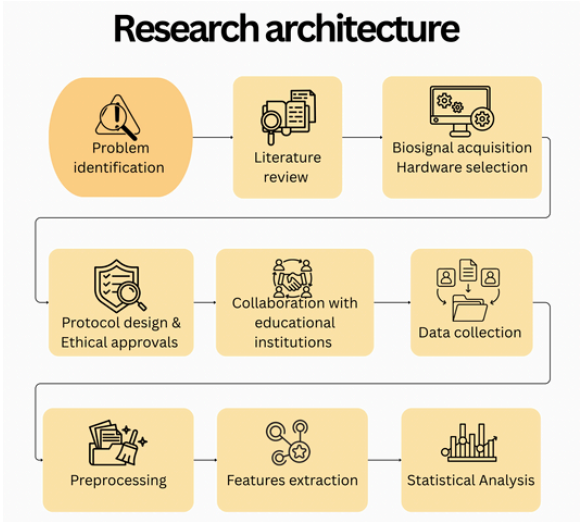
Multi device Syncronization & BioSignal Acquisition

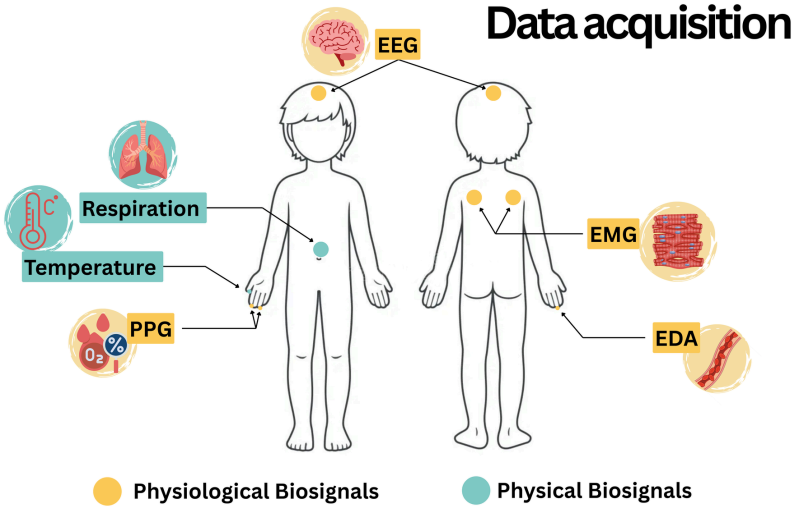


PROJECT 6

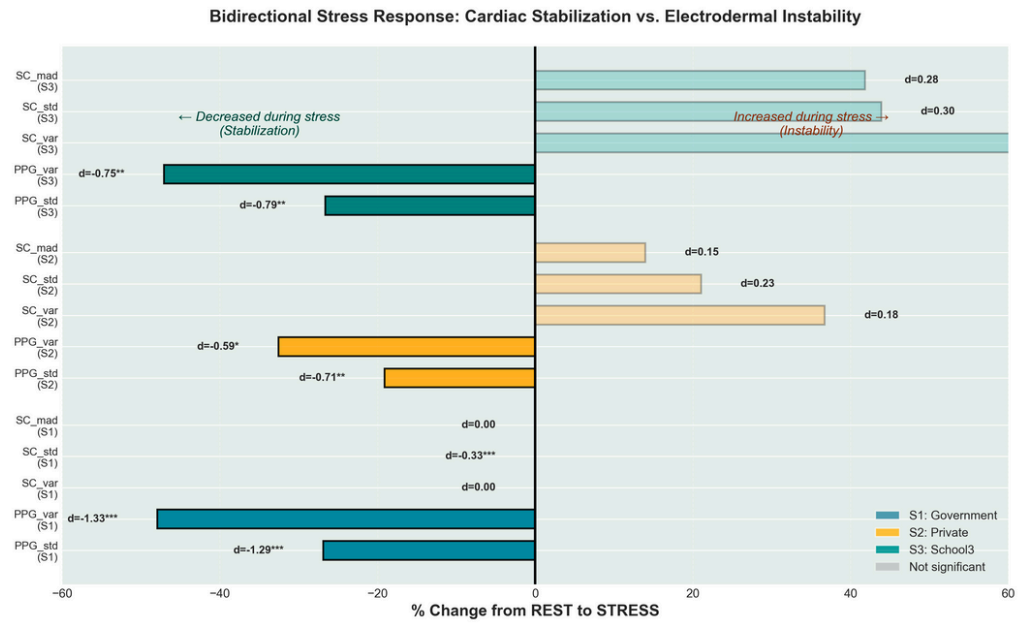
MULTI-INSTITUTIONAL STRESS COMPARISON IN SCHOOL CHILDREN

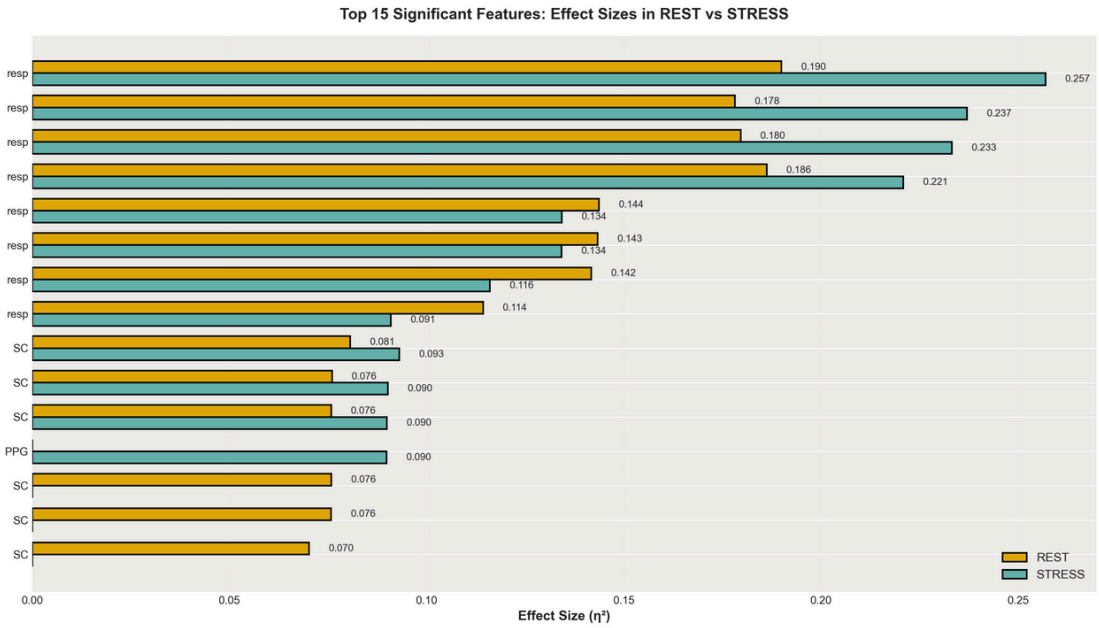
- **Challenge**Limited research exists on measuring stress in children, despite schools being major stressors. Most existing studies rely on only two or three biosignals, leaving gaps in comprehensive stress assessment.
- **Action:** Coordinated across three institutions for acquisition of six types of bio signal, while meeting both ethical and technical requirements. Secured approvals, designed protocols, led data collection, and applied signal processing followed by statistical analysis to bio signals for comparison of stress among institutions. Trained junior researchers on how to conduct data collection.
- **Result:** Produced statically significant results showing highest stress among children from middle class families, strengthened institutional collaboration, and expanded research capacity.

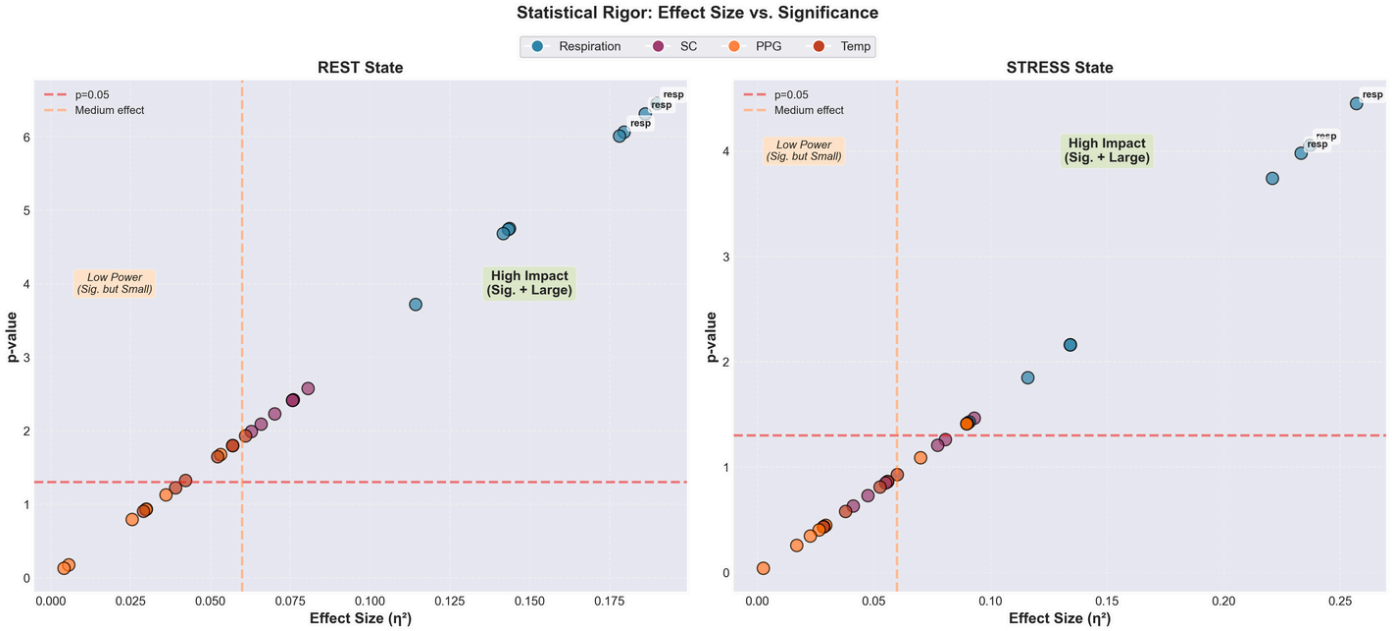


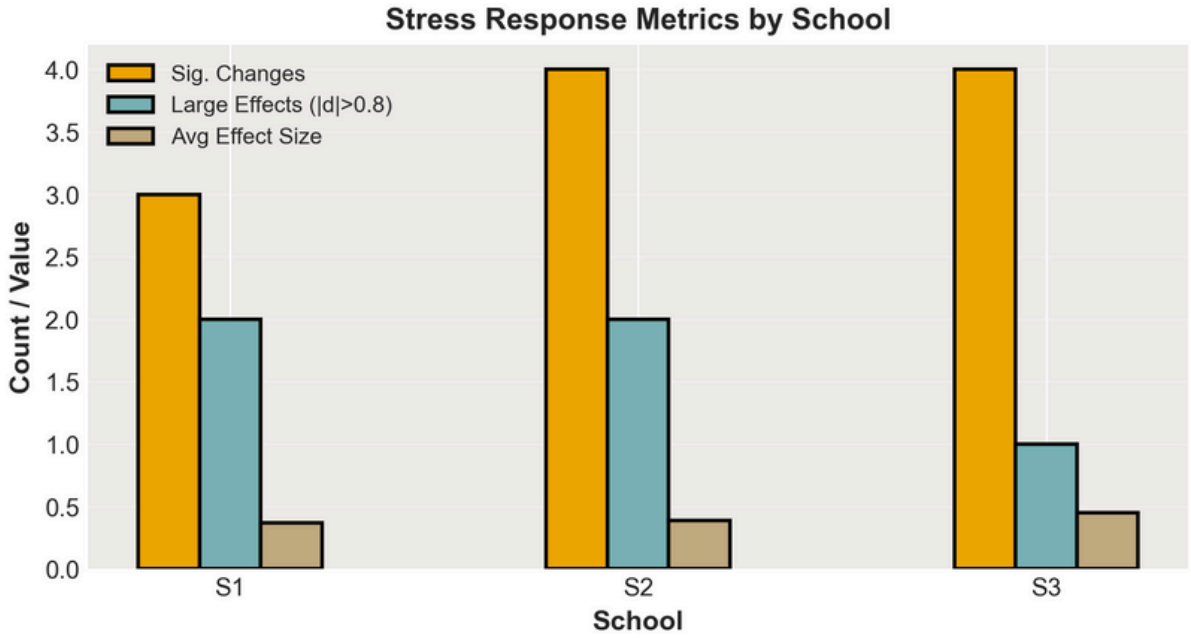







Top 15 Significant Features: Effect Sizes in REST vs STRESS
resp
resp
resp
resp
resp
resp
resp
resp
SC
SC
SC
PPG
SC
SC
SC
0.190
0.257
0.178
0.237
0.180
0.233
0.186
0.221
0.144
0.134
0.143
0.134
0.142
0.116
0.114
0.091
0.081
0.093
0.076
0.090
0.076
0.090
0.090
0.076
0.076
0.070
REST
STRESS
0.00
0.05
0.10
0.15
0.20
0.25
Effect Size (η²)

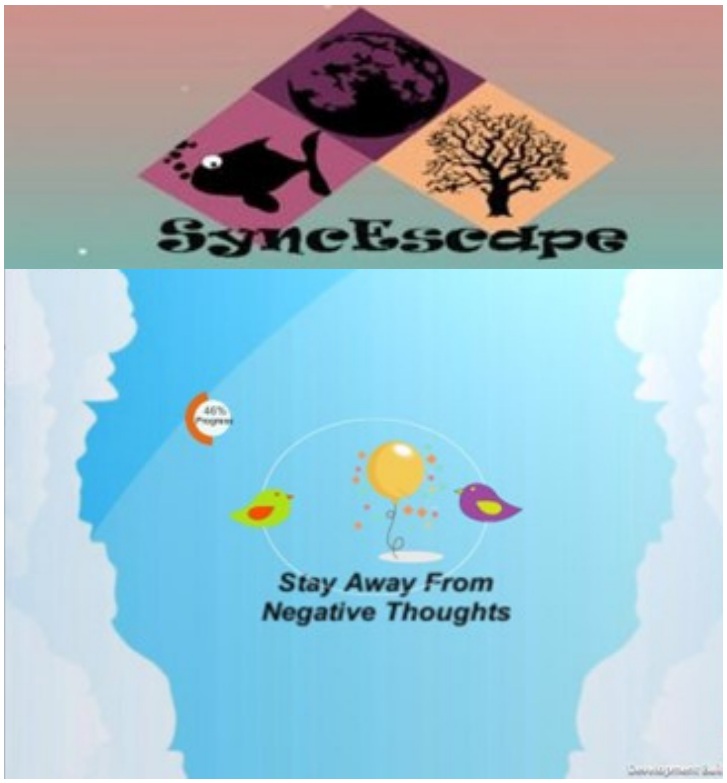

Statistical Rigor: Effect Size vs. Significance
Respiration
SC
PPG
Temp
REST State
STRESS State
p=0.05
Medium effect
Low Power
(Sig. but Small)
High Impact
(Sig. + Large)
resp
p-value
Effect Size (η²)
0.000
0.025
0.050
0.075
0.100
0.125
0.150
0.175
0.00
0.05
0.10
0.15
0.20
0.25

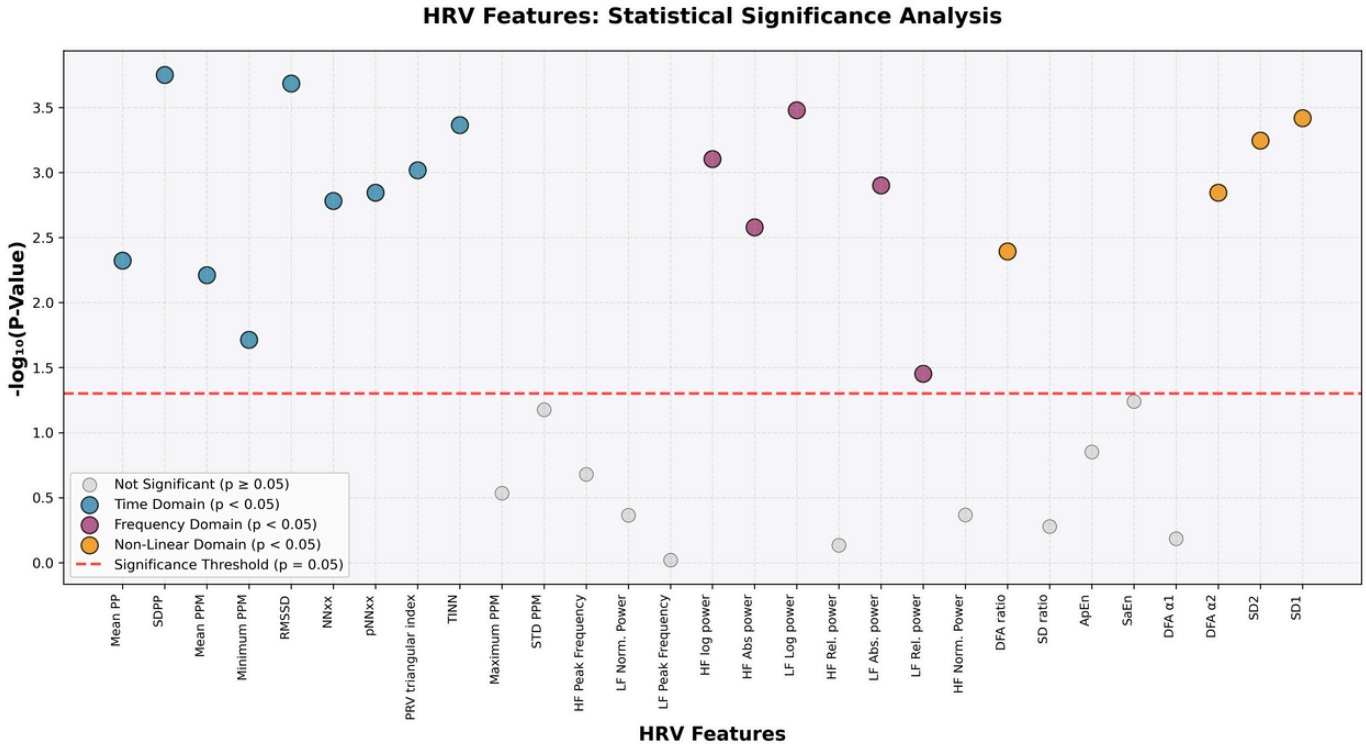

Stress Response Metrics by School
Sig. Changes
Large Effects (|d|>0.8)
Avg Effect Size
Count / Value
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
S1
S2
S3
School

PROJECT 7

BIOMEDICAL ASSESSMENT OF SERIOUS GAMES FOR STRESS MANAGEMENT

- **Challenge:** A software engineer developed SyncEscape, a serious game for stress reduction, but lacked objective physiological validation. Self-reported measures alone couldn't provide scientific credibility for the game's therapeutic effectiveness.
- **Action:** Designed and implemented a PPG-based physiological monitoring system to objectively measure stress during gameplay. Acquired signals from participants in controlled sessions, developed a signal processing pipeline for cardiovascular feature extraction and heart rate variability analysis, and conducted with and without gaming elements game stress level comparisons to validate efficiency of game.
- **Result:** Demonstrated statistically significant stress reduction through controlled experimentation, validating serious gaming as an effective mental health intervention.



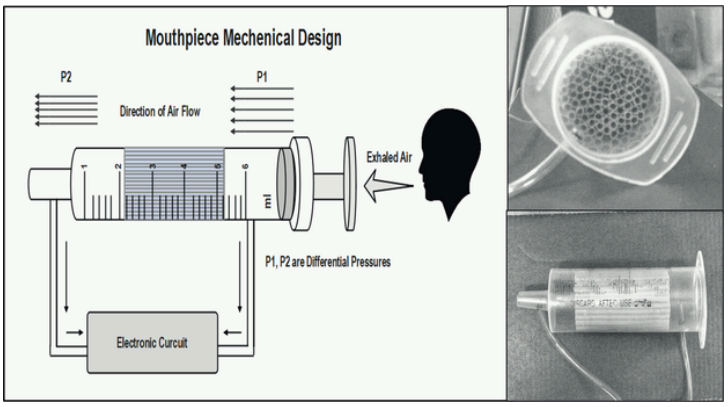


PROJECT 8

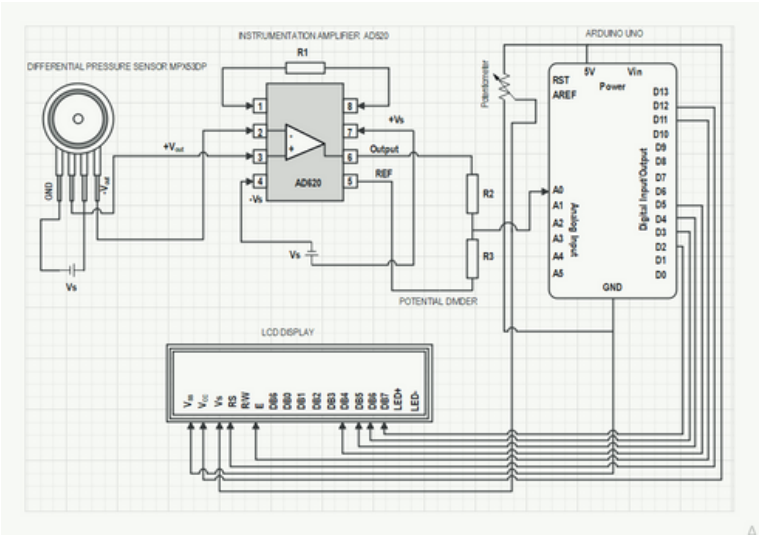
COST-EFFECTIVE SPIROMETER PROTOTYPE

- **Challenge:** Healthcare systems in resource-limited settings required affordable diagnostic solutions, without sacrificing effectiveness. Traditional spirometers are $10,000+ inaccessible to limited budget clinics, creating critical gaps in respiratory disease detection and management.
- **Action:** Designed mechanical, electronic, and coding components, iterated multiple structures to reduce cost, and aligned design with clinical needs.
- **Result:** Produced validated low-cost prototypes, recognized at a research fair for innovation, showcasing persistence and creativity.

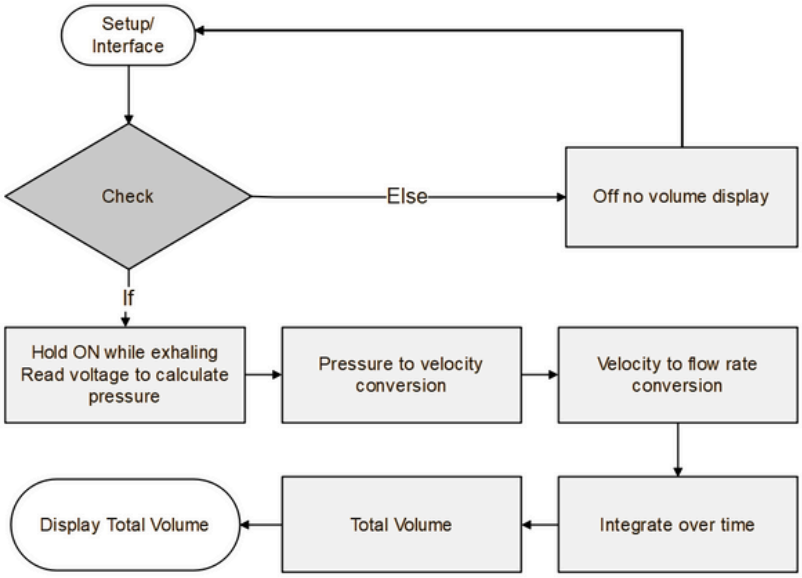
MECHANICAL DESIGN



HARDWARE CONFIGURATION



ALGORITHMIC FLOW FOR VOLUME ESTIMATION



PROJECT 9

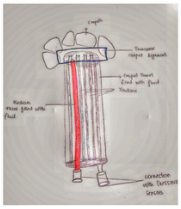
ERGONOMIC MONITORING FOR CARPAL TUNNEL PREVENTION



PROTOTYPE DESIGN

- **Challenge:** Addressed the critical need for ergonomic assessment by calculating wrist flexion and extension angles to prevent carpal tunnel syndrome in workplace and daily activities.
- **Action:** Led a team of three engineers in developing a carpal tunnel prevention prototype, utilizing PASCO software to measure fluid pressure on the median nerve and ensure clinical relevance of the solution.
- **Result:** Developed a prototype solution that identified optimal wrist angles for minimizing median nerve pressure, presented findings at a mock conference, and received recognition for teamwork and clinical impact.

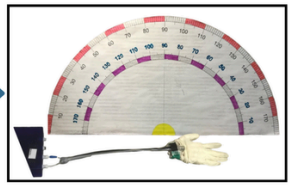
Median Nerve Pressure Detection



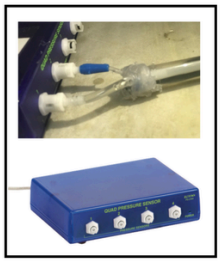
Mechanics Model



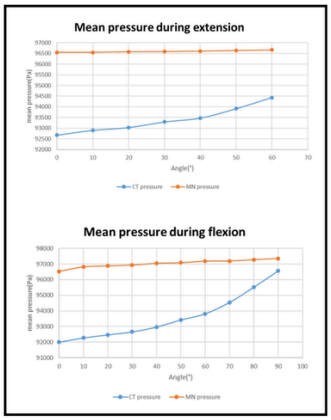
Fluid Mechanics Structure



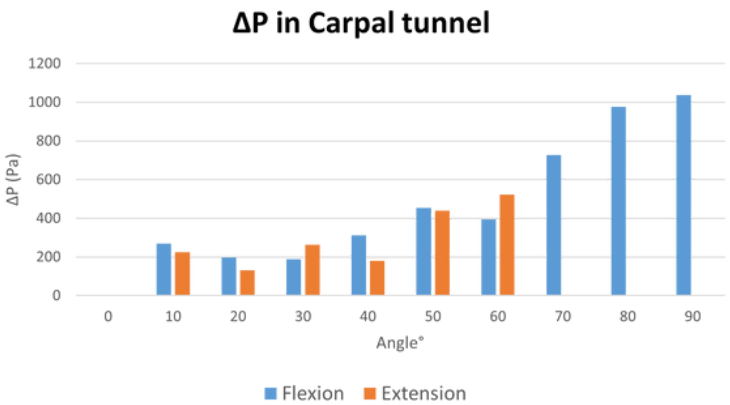
System Integration

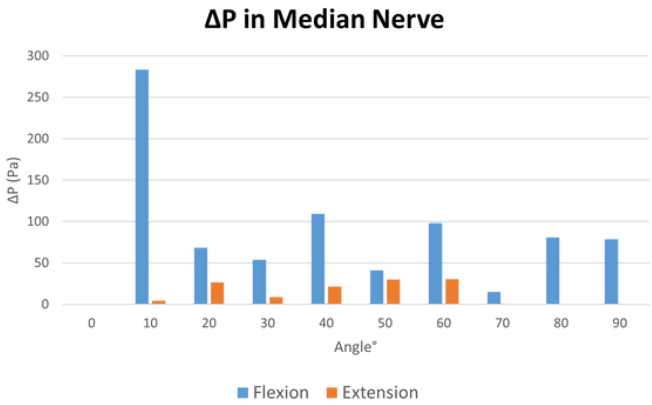


Measurement System
PASPORT Quad Pressure Sensor - PS-2164



Mean pressure of flexion and extension

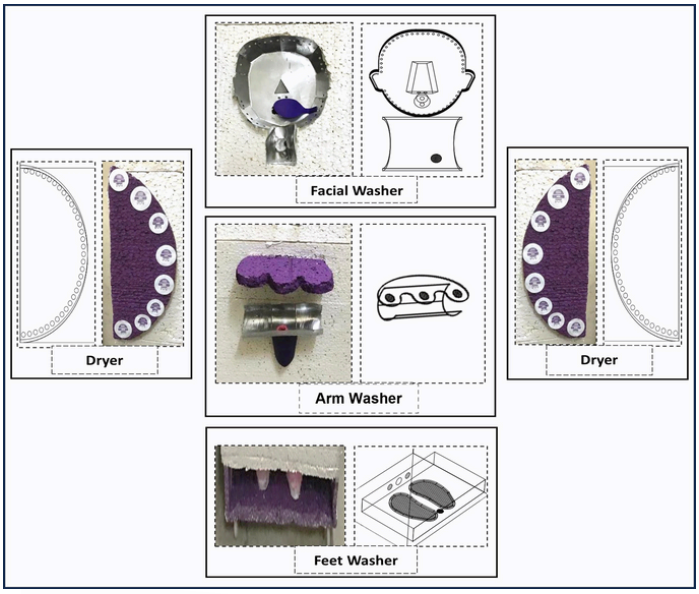




PROJECT 10

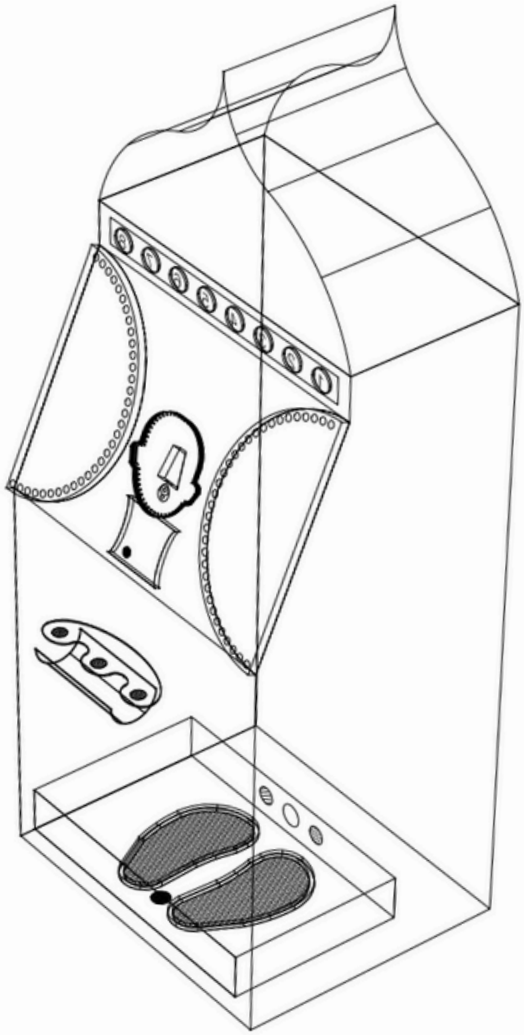
AUTOMATED ABLUTION DEVICE FOR DISABLED MECHANICAL DESIGN AND CROSS-DISCIPLINARY COLLABORATION

- **Challenge:** Needed to design a scalable device prototype for clinical application.
- **Action:** With a team of three engineers worked to design the mechanical structure and to build a human-scale prototype, collaborated with a chemical engineer for 3D AutoCAD modeling, and validated design feasibility.
- **Result:** Demonstrated clinical feasibility, providing empowerment to disabled and strengthened the cross-disciplinary collaboration.



PROTOTYPE DESIGN





AUTOCAD MECHANICAL STRUCTURE

PROJECT 11

3D PROSTHETIC FINGER DESIGN IN SOLIDWORKS



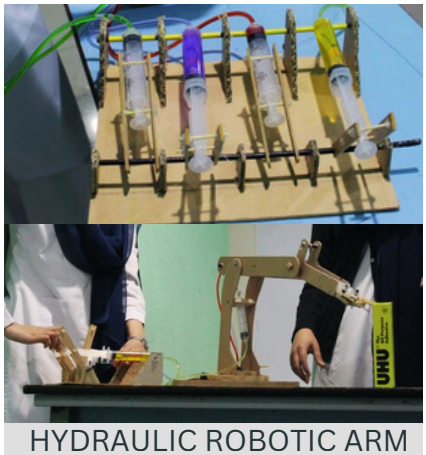
3D PROSTHETIC FINGER DESIGN IN SOLIDWORKS

- **Challenge:** To advance prosthetic hand development, an accurate 3D biomechanical model of a human finger was needed to replicate natural finger physiology and movement.
- **Action:** Designed a complete anatomically accurate finger in SolidWorks with three phalanges, realistic joints (MCP, PIP, DIP), and screw-based connectors. Used advanced CAD techniques, including surface extrusion, revolve features, and trim operations to replicate natural finger structure and curvature
- **Result:** Created a fully functional 3D finger model with realistic biomechanics and modular design, providing a scalable foundation for prosthetic hand system development.

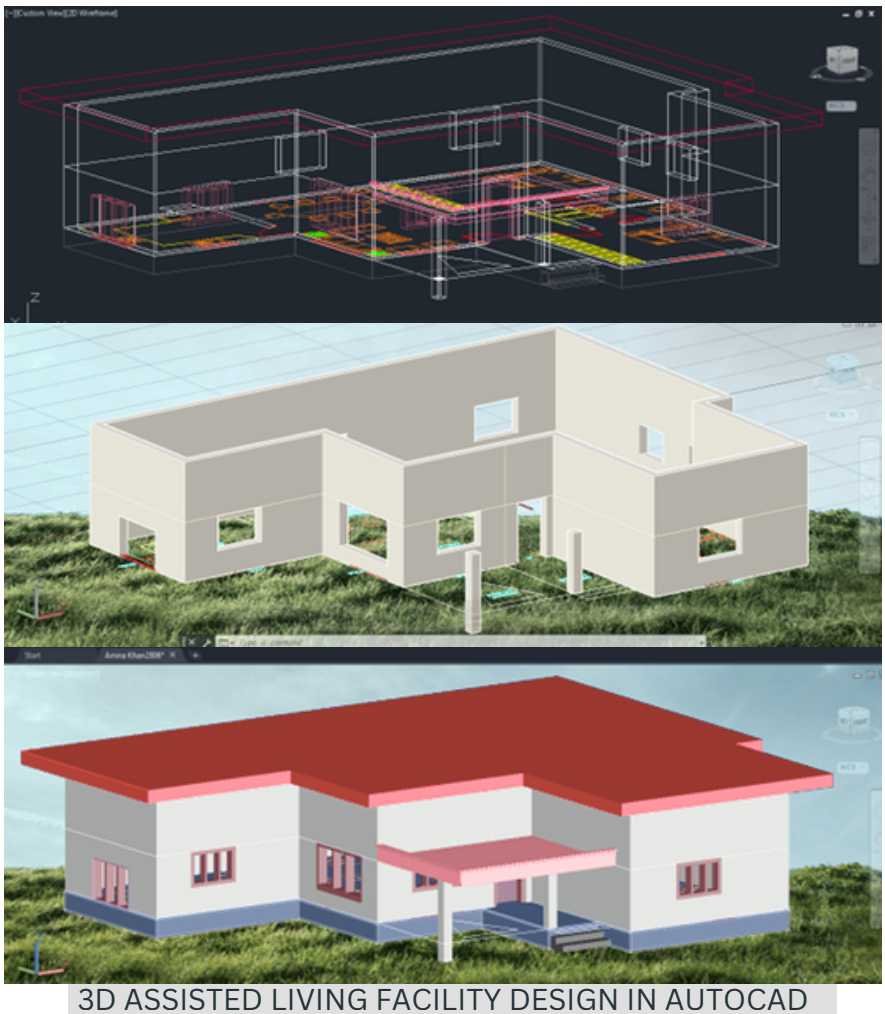
PROJECT HIGHLIGHTS



MOLD MAKING & CASTING HIP IMPLANT PROTOTYPE



HYDRAULIC ROBOTIC ARM



3D ASSISTED LIVING FACILITY DESIGN IN AUTOCAD